

Администрирование сетевых подсистем

Настройка файловых служб Samba

Талебу Тенке Франк Устон

Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Цель лабораторной работы

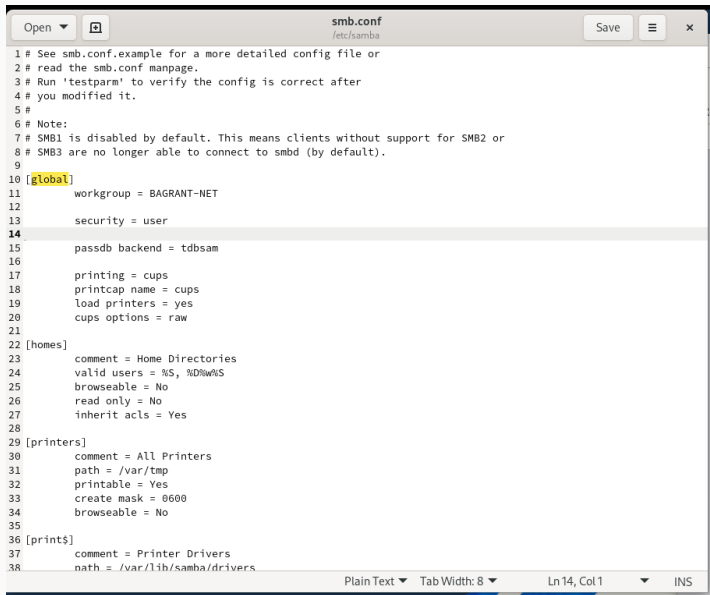
Приобретение практических навыков настройки доступа групп пользователей к общим ресурсам по протоколу SMB (Samba).

Выполнение лабораторной работы

Настройка сервера Samba

```
root@server:~  
agrant@server ~]$ dnf -y install samba samba-client cifs-utils  
Error: This command has to be run with superuser privileges (under the root user on most systems).  
agrant@server ~]$ sudo -i  
root@server ~]# dnf -y install samba samba-client cifs-utils  
Package Architecture Version Repository Size  
Installing:  
cifs-utils x86_64 7.2-1.el9 baseos 113 k  
samba x86_64 4.22.4-12.el9_7 baseos 930 k  
samba-client x86_64 4.22.4-12.el9_7 appstream 760 k  
Installing dependencies:  
libnetapi x86_64 4.22.4-12.el9_7 baseos 142 k  
samba-common-tools x86_64 4.22.4-12.el9_7 baseos 483 k  
samba-dcerpc x86_64 4.22.4-12.el9_7 baseos 717 k  
samba-ldb-ldap-modules x86_64 4.22.4-12.el9_7 baseos 34 k  
samba-libs x86_64 4.22.4-12.el9_7 baseos 123 k  
  
Transaction Summary  
-----  
Install 8 Packages  
  
Total download size: 3.2 M  
Installed size: 12 M
```

Настройка конфигурации Samba



```
1 # See smb.conf.example for a more detailed config file or
2 # read the smb.conf manpage.
3 # Run 'testparm' to verify the config is correct after
4 # you modified it.
5 #
6 # Note:
7 # SMB1 is disabled by default. This means clients without support for SMB2 or
8 # SMB3 are no longer able to connect to smbd (by default).
9
10 [global]
11     workgroup = BAGRANT-NET
12
13     security = user
14
15     passdb backend = tdbsam
16
17     printing = cups
18     printcap name = cups
19     load printers = yes
20     cups options = raw
21
22 [homes]
23     comment = Home Directories
24     valid users = %S, %D%w%S
25     browseable = No
26     read only = No
27     inherit acls = Yes
28
29 [printers]
30     comment = All Printers
31     path = /var/tmp
32     printable = Yes
33     create mask = 0600
34     browseable = No
35
36 [print$]
37     comment = Printer Drivers
38     path = /var/lib/samba/drivers
```

Plain Text ▾ Tab Width: 8 ▾ Ln 14, Col 1 ▾ INS

Запуск и проверка службы Samba

```
Complete!
[root@server ~]# groupadd -g 1010 sambagroup
[root@server ~]# usermod -aG sambagroup user
[root@server ~]# mkdir -p /srv/smbashare
[root@server ~]# gedit /etc/samba/smb.conf

(gedit:9023): dconf-WARNING **: 15:20:19.112: failed to commit changes to dconf: Failed
to execute child process "dbus-launch" (No such file or directory)

(gedit:9023): dconf-WARNING **: 15:20:19.121: failed to commit changes to dconf: Failed
to execute child process "dbus-launch" (No such file or directory)

(gedit:9023): dconf-WARNING **: 15:20:19.620: failed to commit changes to dconf: Failed
to execute child process "dbus-launch" (No such file or directory)

(gedit:9023): dconf-WARNING **: 15:20:19.620: failed to commit changes to dconf: Failed
to execute child process "dbus-launch" (No such file or directory)
```

Рис. 3: Проверка работы службы Samba

Настройка межсетевого экрана

```
35 [print$]
36     comment = Printer Drivers
37     path = /var/lib/samba/drivers
38     # printadmin is a local group
39     write list = printadmin root
40     force group = printadmin
41     create mask = 0664
42     directory mask = 0775
43 [sambashare]
44     comment = My Samba Share
45     path = /srv/sambashare
46     write list = @sambagroup
```

Plain Text ▼ Tab Width: 8 ▼ Ln 44, Col 25 ▼ INS

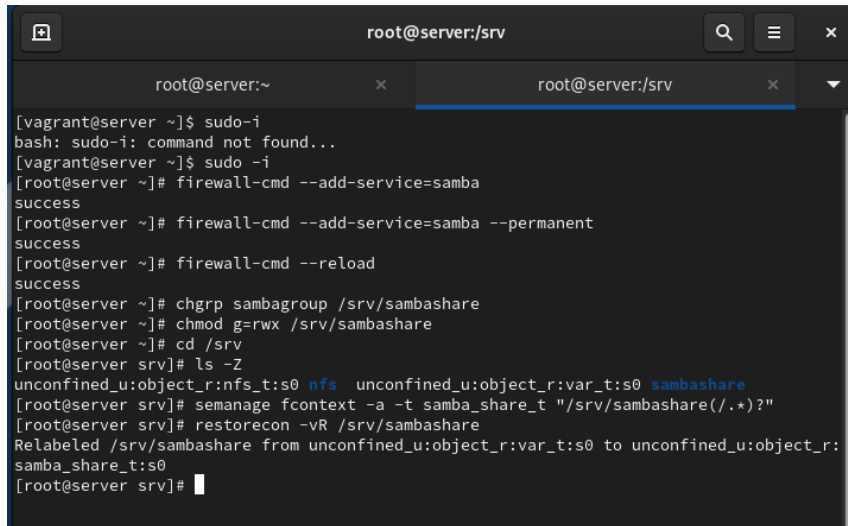
Рис. 4: Просмотр файла samba.xml

Настройка прав доступа с SELinux

```
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>Samba</short>
  <description>This option allows you to access and participate in Windows file and printer sharing networks. You need the samba package installed for this option to be useful.</description>
  <include service="samba-client"/>
  <port protocol="tcp" port="139"/>
  <port protocol="tcp" port="445"/>
</service>
/usr/lib/firewalld/services/samba.xml (END)
```

Рис. 5: Настройка firewalld, прав доступа с SELinux

Проверка контекста SELinux



The image shows a terminal window with a dark theme. The title bar at the top reads 'root@server:/srv'. Below the title bar, there are two tabs: 'root@server:~' and 'root@server:/srv', with the second tab being active. The terminal content shows a user switching from 'vagrant' to 'root' using 'sudo -i'. Then, they run 'firewall-cmd' to add the 'samba' service and reload the firewall. Next, they create a group 'sambagroup' for '/srv/sambashare' and set permissions 'g=rwx'. Finally, they use 'semanage' to set the context to 'samba_share_t' and 'restorecon' to apply it. The output shows the context was relabeled from 'unconfined_u:object_r:nfs_t:s0' to 'unconfined_u:object_r:var_t:s0 samba_share_t:s0'.

```
root@server:/srv

[vagrant@server ~]$ sudo-i
bash: sudo-i: command not found...
[vagrant@server ~]$ sudo -i
[root@server ~]# firewall-cmd --add-service=samba
success
[root@server ~]# firewall-cmd --add-service=samba --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# chgrp sambagroup /srv/sambashare
[root@server ~]# chmod g=rwx /srv/sambashare
[root@server ~]# cd /srv
[root@server srv]# ls -Z
unconfined_u:object_r:nfs_t:s0 nfs unconfined_u:object_r:var_t:s0 sambashare
[root@server srv]# semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
[root@server srv]# restorecon -vR /srv/sambashare
Relabeled /srv/sambashare from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:
samba_share_t:s0
[root@server srv]#
```

Рис. 6: Проверка контекста SELinux

Проверка групп пользователя

```
root@client:~  
[vagrant@client ~]$ sudo -i  
[root@client ~]# dnf -y install samba-client cifs-utils  
Rocky Linux 9 - BaseOS           8.6 kB/s | 4.3 kB    00:00  
Rocky Linux 9 - BaseOS           6.5 MB/s | 16 MB     00:02  
Rocky Linux 9 - AppStream        15 kB/s | 4.8 kB     00:00  
Rocky Linux 9 - AppStream        4.4 MB/s | 17 MB     00:03  
Rocky Linux 9 - Extras           7.3 kB/s | 3.1 kB     00:00  
Rocky Linux 9 - Extras          33 kB/s | 17 kB     00:00  
Dependencies resolved.  
=====
```

Package	Arch	Version	Repository	Size
---------	------	---------	------------	------

```
=====
```

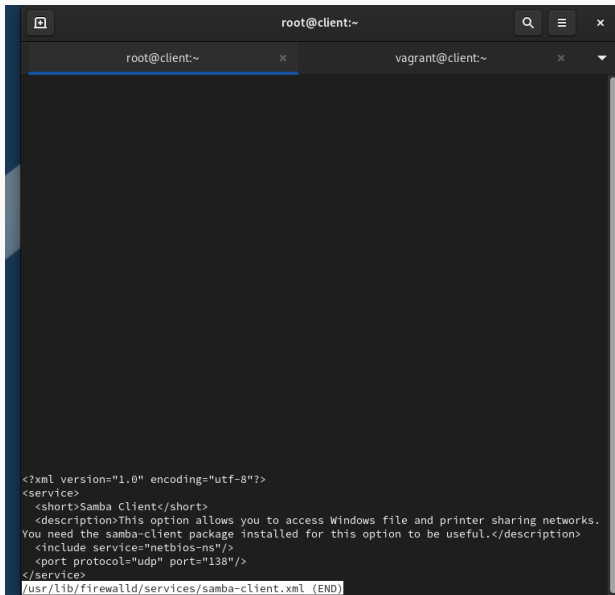
Installing:

cifs-utils	x86_64	7.2-1.el9	baseos	113 k
samba-client	x86_64	4.22.4-12.el9_7	appstream	760 k

Upgrading:

libipa_hbac	x86_64	2.9.7-4.el9_7.1	baseos	33 k
libldb	x86_64	4.22.4-12.el9_7	baseos	180 k
libsmbclient	x86_64	4.22.4-12.el9_7	baseos	73 k
libsss_certmap	x86_64	2.9.7-4.el9_7.1	baseos	88 k
libsss_idmap	x86_64	2.9.7-4.el9_7.1	baseos	39 k
libsss_nss_idmap	x86_64	2.9.7-4.el9_7.1	baseos	43 k
libsss_sudo	x86_64	2.9.7-4.el9_7.1	baseos	33 k
libtalloc	x86_64	2.4.3-1.el9	baseos	33 k
libtdb	x86_64	1.4.13-1.el9	baseos	53 k
libtevent	x86_64	0.16.2-1.el9	baseos	50 k
libwbclient	x86_64	4.22.4-12.el9_7	baseos	42 k
samba-client-libs	x86_64	4.22.4-12.el9_7	baseos	5.3 M
samba-common	noarch	4.22.4-12.el9_7	baseos	173 k
samba-common-libs	x86_64	4.22.4-12.el9_7	baseos	104 k
sssd	x86_64	2.9.7-4.el9_7.1	baseos	25 k
sssd-ad	x86_64	2.9.7-4.el9_7.1	baseos	217 k
sssd-client	x86_64	2.9.7-4.el9_7.1	baseos	158 k
sssd-common	x86_64	2.9.7-4.el9_7.1	baseos	1.6 M
sssd-common-pac	x86_64	2.9.7-4.el9_7.1	baseos	94 k
sssd-ipa	x86_64	2.9.7-4.el9_7.1	baseos	286 k
sssd-kcm	x86_64	2.9.7-4.el9_7.1	baseos	107 k
sssd-krb5	x86_64	2.9.7-4.el9_7.1	baseos	70 k

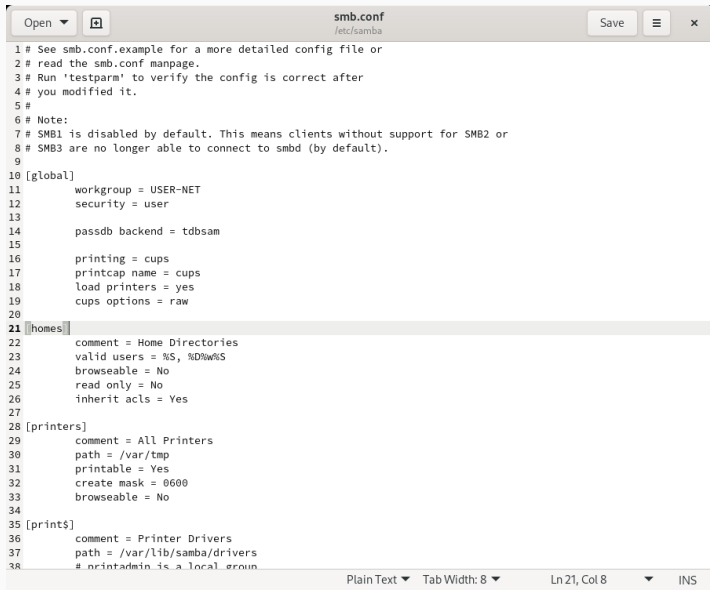
Проверка записи em общий каталог



A terminal window with a dark background and light blue text. The window title bar shows 'root@client:~' and search, menu, and close icons. Two tabs are open: 'root@client:~' (active) and 'vagrant@client:~'. The terminal content displays an XML configuration snippet for a Samba Client service, including a description and an include statement for netbios-ns. The path '/usr/lib/firewalld/services/samba-client.xml (END)' is highlighted at the bottom.

```
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>Samba Client</short>
  <description>This option allows you to access Windows file and printer sharing networks.
You need the samba-client package installed for this option to be useful.</description>
  <include service="netbios-ns"/>
  <port protocol="udp" port="138"/>
</service>
/usr/lib/firewalld/services/samba-client.xml (END)
```

Просмотр ресурсов сервера с клиента



```
1 # See smb.conf.example for a more detailed config file or
2 # read the smb.conf manpage.
3 # Run 'testparm' to verify the config is correct after
4 # you modified it.
5 #
6 # Note:
7 # SMB1 is disabled by default. This means clients without support for SMB2 or
8 # SMB3 are no longer able to connect to smbd (by default).
9
10 [global]
11     workgroup = USER-NET
12     security = user
13
14     passdb backend = tdbsam
15
16     printing = cups
17     printcap name = cups
18     load printers = yes
19     cups options = raw
20
21 [homes]
22     comment = Home Directories
23     valid users = %S, %D\\%S
24     browseable = No
25     read only = No
26     inherit acls = Yes
27
28 [printers]
29     comment = All Printers
30     path = /var/tmp
31     printable = Yes
32     create mask = 0600
33     browseable = No
34
35 [print$]
36     comment = Printer Drivers
37     path = /var/lib/samba/drivers
38     # printadmin is a local group
```

Plain Text ▾ Tab Width: 8 ▾ Ln 21, Col 8 ▾ INS

Монтирование общего ресурса

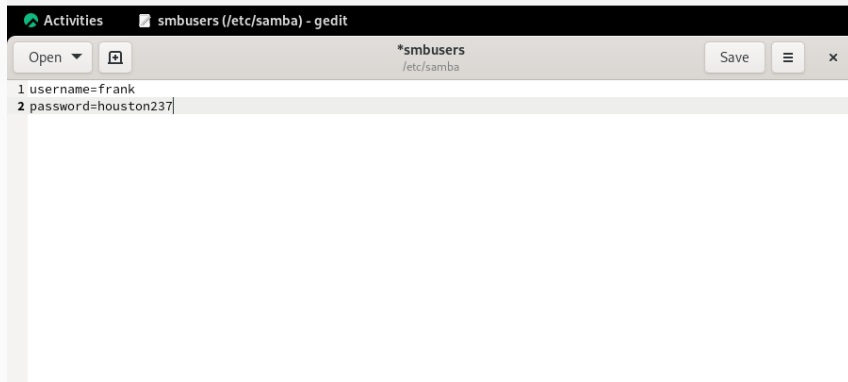


Рис. 10: Монтирование общего ресурса Samba

Проверка записи по клиенте

```
10 # units generated from this file.
11 #
12 UUID=c84cce45-9089-48d9-9617-2f1d1bd45fdd /          xfs      defaults    0 0
13 /swapfile none swap defaults 0 0
14 #VAGRANT-BEGIN
15 # The contents below are automatically generated by Vagrant. Do not modify.
16 vagrant /vagrant vboxsf uid=1000,gid=1000,_netdev 0 0
17 #VAGRANT-END
18
19 /server/sambashare /mnt/samba cifs
20 user,rw,uid=user_name,gid=sambagroup,
21 credentials=/etc/samba/smbusers,_netdev 0 0|
```

Plain Text ▼

Tab Width: 8 ▼

Ln 21, Col 44 ▼

INS

Рис. 11: Проверка записи em смонтированный ресурс

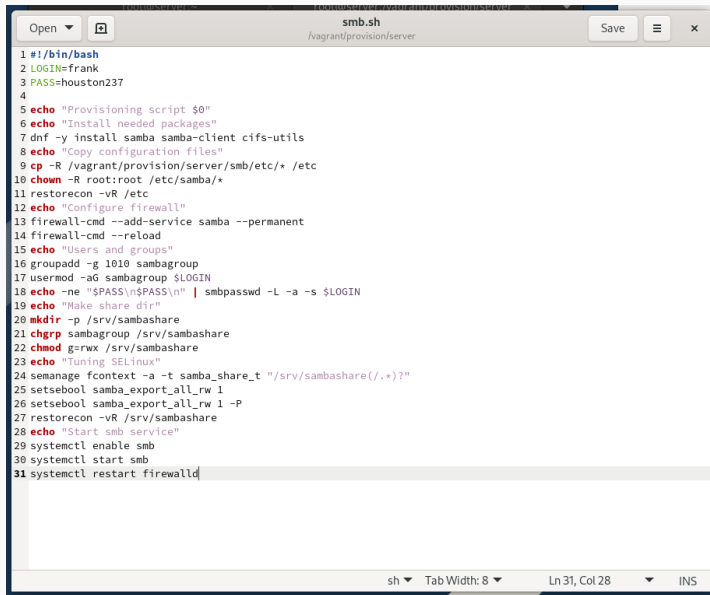
Файл учётных данных Samba

```
Added user user.
[root@server sambashare]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/smb/etc/samba
[root@server server]# cp -R /etc/samba/smb.conf /vagrant/provision/server/smb/etc/samba
/
[root@server server]# cd /vagrant/provision/server
[root@server server]# touch smb.sh
[root@server server]# chmod +x smb.sh
[root@server server]# gedit smb
smb/      smb.sh
[root@server server]# gedit smb.sh

(gedit:9421): dconf-WARNING **: 16:21:35.722: failed to commit changes to dconf: Failed
to execute child process "dbus-launch" (No such file or directory)
```

Рис. 12: Файл учётных данных smbusers

Автоматическое монтирование



```
smb.sh
/vagrant/provision/server

1 #!/bin/bash
2 LOGIN=frank
3 PASS=houston237
4
5 echo "Provisioning script $0"
6 echo "Install needed packages"
7 dnf -y install samba samba-client cifs-utils
8 echo "Copy configuration files"
9 cp -R /vagrant/provision/server/smb/etc/* /etc
10 chown -R root:root /etc/samba/*
11 restorecon -vR /etc
12 echo "Configure firewall"
13 firewall-cmd --add-service samba --permanent
14 firewall-cmd --reload
15 echo "Users and groups"
16 groupadd -g 1010 sambagroup
17 usermod -aG sambagroup $LOGIN
18 echo -ne "$PASS\n$PASS\n" | smbpasswd -L -a -s $LOGIN
19 echo "Make share dir"
20 mkdir -p /srv/sambashare
21 chgrp sambagroup /srv/sambashare
22 chmod g=rwx /srv/sambashare
23 echo "Tuning SELinux"
24 semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
25 setsebool samba_export_all_rw 1
26 setsebool samba_export_all_rw 1 -P
27 restorecon -vR /srv/sambashare
28 echo "Start smb service"
29 systemctl enable smb
30 systemctl start smb
31 systemctl restart firewalld
```

sh Tab Width: 8 Ln 31, Col 28 INS

Проверка автоматического монтирования

```
smb.sh
/vagrant/provision/server

1 #!/bin/bash
2 LOGIN=frank
3 PASS=houston237
4
5 echo "Provisioning script $0"
6 echo "Install needed packages"
7 dnf -y install samba samba-client cifs-utils
8 echo "Copy configuration files"
9 cp -R /vagrant/provision/server/smb/etc/* /etc
10 chown -R root:root /etc/samba/*
11 restorecon -vR /etc
12 echo "Configure firewall"
13 firewall-cmd --add-service samba --permanent
14 firewall-cmd --reload
15 echo "Users and groups"
16 groupadd -g 1010 sambagroup
17 usermod -aG sambagroup $LOGIN
18 echo -ne "$PASS\n$PASS\n" | smbpasswd -L -a -s $LOGIN
19 echo "Make share dir"
20 mkdir -p /srv/sambashare
21 chgrp sambagroup /srv/sambashare
22 chmod g=rwx /srv/sambashare
23 echo "Tuning SELinux"
24 semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
25 setsebool samba_export_all_rw 1
26 setsebool samba_export_all_rw 1 -P
27 restorecon -vR /srv/sambashare
28 echo "Start smb service"
29 systemctl enable smb
30 systemctl start smb
31 systemctl restart firewalld
```

sh Tab Width: 8 Ln 31, Col 28 INS

Проверка доступа после монтирования

```
(gedit:4874): dconf-WARNING **: 16:21:57.020: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@client samba]# mount -a
mount: /etc/fstab: parse error at line 20 -- ignored
mount.cifs: bad UNC (/server/smbashare)
mount: (hint) your fstab has been modified, but systemd still uses
        the old version; use 'systemctl daemon-reload' to reload.
mount: 0: mount point does not exist.
[root@client samba]# cd /vagrant/provision/client
[root@client client]# mkdir -p /vagrant/provision/client/smb/etc/samba
[root@client client]# cp -R /etc/samba/smb.conf /vagrant/provision/client/smb/etc/samba/
[root@client client]# cp -R /etc/samba/smbusers /vagrant/provision/client/smb/etc/samba/
[root@client client]# cd /vagrant/provision/client
[root@client client]# touch smb.sh
[root@client client]# chmod +x smb.sh
[root@client client]# gedit smb.sh

(gedit:4959): dconf-WARNING **: 16:32:55.868: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:55.874: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:56.084: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:56.110: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:56.110: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
```

Выводы по проделанной работе

В ходе лабораторной работы выполнена настройка сервера и клиента Samba в виртуальной среде.

Реализован общий ресурс с разграничением прав доступа на основе групп Linux и механизмов SELinux.

Проверена работа аутентифицированного доступа и автоматического монтирования ресурсов.

Настройка автоматизирована с использованием Vagrant.